Inyoung Oh

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Fusing semantic and geometric cues for reliable, metric 3D spatial perception.

Research Focus

My research makes learned and semantic 3D *metric, consistent, and structured* by fusing it with explicit geometry, recovering reliable spatial structure exactly where data-driven models are least faithful: at object boundaries, sharp discontinuities, and absolute scale.

- Scene-intrinsic geometric cues (surface normals, sharp boundaries, ground planes) as transferable structure injected into learned and semantic 3D
- Metric, consistent 3D from a single image: bridging 2D and 3D through scene geometry, where learned depth and multi-view reconstruction fall short
- Geometric representations that generalize across sensors and domains (CAD, LiDAR, indoor/outdoor scans, monocular)

Education

Ph.D., Mechanical and Robotics Engineering, Gwangju Institute of Science and Technology (GIST) **Feb 2026**
Dissertation: *Normal Vector Estimation, and Semantic Segmentation of 3D Point Clouds using Deep Learning and Geometric Analysis*

M.S., Mechatronics, GIST **Feb 2018**
Thesis: *Sphere and Cylinder Detection in Kinect Point Clouds using RANSAC and the 2D Hough Transform*

B.S., Mechatronics, Chungnam National University **Feb 2016**

Research Experience

Postdoctoral Fellow, Center for AI, KIST | Advisor: Dr. Junghyun Cho **Feb 2026–Present**

- Reframed metric relocalization from a moving camera as a **geometric 2D–3D bridge**: on an up-to-scale map, metric placement reduces to a ray–ground intersection set by one scene-intrinsic quantity, the ground-plane normal, independent of pose and reconstruction method.
- The geometric estimator recovers metric position **7.7–100**× better than monocular-depth and VGGT-style feed-forward baselines, even given ground-truth pose, across indoor and outdoor capture (in preparation [P1]).

Research Assistant, Modeling and Simulation Lab, GIST | Advisor: Prof. Kwang Hee Ko **Feb 2016–Feb 2026**

Established the geometric half of this fusion: explicit scene-intrinsic cues (surface normals, sharp features), injected into learned models, that transfer across sensors, sampling density, and noise.

Sharp Feature Detection & Geometric Descriptors

- **SFD-Net** (ECCV 2026): an architecture-agnostic descriptor detecting sharp features as surface-normal discontinuities; prepended to three backbones *unmodified* it adds +4–7 pp Average F1, transfers *zero-shot* from 10K-point CAD to scans ∼36× larger, and is SOTA on ABC across all four noise conditions. [C1]
- Replacing its heuristic scales with a training-free directional-statistics prior that stays informative at discontinuities (in preparation). [P2]

Normal Vector Estimation

- Multi-scale normal estimator with attention-based scale weighting and Chamfer-distance noise-aware supervision; SOTA on the PCPNet primitive subset (noise-free RMSE 1.821°). [J1]

Semantic Segmentation (Outdoor & Indoor)

- Injected intensity-assisted normals into RandLA-Net via a Normal Local Feature Aggregation module: +4.0 pp mIoU on SemanticKITTI (53.9→57.9), with the largest gains on safety-critical classes (e.g. Bicycle +21.4 pp). [J3]
- Extended refined normals into boundary-aware indoor segmentation: +6.3% mIoU over the PushBoundary baseline on S3DIS (Ph.D. thesis).

6-DoF Pose Estimation & MR Systems

- Designed RoI-PCA color augmentation decoupling object appearance from background: +10.77 pp 5cm5° accuracy on LINEMOD; integrated into a real-time, markerless multi-device MR collaboration system. [J2]

Geometric Primitive Recognition

- Prior-free pipeline recognition from raw point clouds (curvature-driven classification + sphere-RANSAC centerline recovery), with no user priors for radius, orientation, or connectivity. [J4]

Publications

Published / Accepted

[C1] **Inyoung Oh**; K.H. Ko. “SFD-Net: Sharp Feature Detection Network Based on Local Geometric Features.” *European Conference on Computer Vision (ECCV)*, **2026**. *(accepted)* [project] [code]

[C2] T.M. Bui; D. Min; **Inyoung Oh**; J. Cho. "Lighting-Aware Diffusion-based Data Augmentation for Robust Low-Light Re-Identification." *British Machine Vision Conference (BMVC)*, **2026**. *(accepted)*

[J1] **Inyoung Oh**; J. Song; M. Kim; D. Yun; K.H. Ko. "Attention-Guided Multi-Scale Point Cloud Normal Estimation with Noise-Aware Training." *Int. J. Image and Graphics*, **2026**. *(accepted)*

[J2] **Inyoung Oh**; G. Jang; J. Song; M. Son; D. Kim; J. Yun; K.H. Ko. "A Mixed Reality-based Remote Collaboration Framework Using Improved Pose Estimation." *Computers in Industry*, 174, 104414, **2026**.

[J3] M. Kim; **Inyoung Oh**; D. Yun; K.H. Ko. "Improved Semantic Segmentation Network Using Normal Vector Guidance for LiDAR Point Clouds." *J. Computational Design and Engineering*, 10(6), 2332–2344, **2023**.

[J4] **Inyoung Oh**; K.H. Ko. "Automated Recognition of 3D Pipelines from Point Clouds." *The Visual Computer*, 37(6), 1385–1400, **2021**.

In Preparation

[P1] **Inyoung Oh** et al. "Metric Relocalization in Up-to-Scale Maps via Ground-Plane Geometry."

[P2] **Inyoung Oh** et al. "A Directional Statistics Approach to Geometric Sharpness Estimation in 3D Point Clouds."

Patent

K.H. Ko; **Inyoung Oh**. "Automated System for Detecting 3D Pipeline." KR 2010-2135828 B1 (Registered), **July 2020**.

Awards & Scholarships

Best Poster Award (×5), Korean Society for Computational Design and Engineering (CDE)	**2020–2025**
CDE DX Encouragement Award, Korean CDE Society	**2022**
Outstanding PhD Student RA Scholarship (×6), GIST	**2018–2025**
GIST Full Scholarship (Government-funded, Doctoral & Master's)	**2016–2026**

Invited Seminars & Presentations

Invited seminar, KEPCO Research Institute: "Object-detection-based 3D LiDAR intensity calibration and real-time detection of clustered object candidates."	**2021**
Inyoung Oh; K.H. Ko. "Automatic Detection of Cylindrical Objects from Unorganized Point Cloud." *EuroVR* (poster), London, UK.	**2018**
Inyoung Oh; K.H. Ko. "A RANSAC-based Method for Detection of Multiple Spheres from a Point Cloud." *Computer Graphics International* (poster), Yokohama, Japan.	**2017**

Teaching & Mentorship

Mentorship, Modeling and Simulation Lab, GIST: mentored M. Kim from undergraduate intern through graduate studies; mentee published a first-author journal paper (*JCDE*, 2023). Also guided additional interns and supervised a bachelor's thesis.	**2018–2026**
Teaching Assistant, Engineering Analysis, GIST: recitations, assessment design, Python/MATLAB/C++ instruction.	**2017, 2020**

Professional Service

Reviewer, *Journal of Computational Design and Engineering* (JCDE)

Technical Skills

Languages & Frameworks: Python, PyTorch, C++, MATLAB, Open3D, PCL, Unity3D
Sensors & Devices: Velodyne/Ouster LiDAR, Kinect v2, Azure Kinect, MR smart glasses

References

Prof. Kwang Hee Ko	**Prof. Bochang Moon**	**Prof. Jinho Song**
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